

Competitive Exam Practice 5 — Answer Key

Addressing Modes, CPU Organization, and Bus Structures

PCC CS-401: Computer Organization and Architecture

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- Q1 (C) Displacement (indexed).** $EA = [\text{register}] + d$ is recomputed from a register on every execution, so it correctly tracks a changing loop index; direct addressing's fixed field cannot.
- Q2 (B) B .** Indirect addressing: $EA = M[A] = B$. The operand itself (42) is found at $M[B]$, but the *effective address* is B , not the operand's value.
- Q3 200.** $EA = [R4] + d = 175 + 25 = 200$.
- Q4 (C) 3.** One step to stage the first operand into a buffer register, one to bring the second operand and compute (result into a second buffer register), one to write the result back — the structural minimum when only one value can occupy the bus per cycle.
- Q5 (C).** Because Bus A and Bus B are separate physical paths and Bus C does not compete with either, both operand reads and the result write-back can all occur in a single clock cycle.
- Q6 2.** One memory access to fetch the pointer from $M[A]$, one to fetch the operand from $M[\text{pointer value}]$.